

Justify your answers and show your work. Unless stated otherwise, all problems are worth 5 points.

Evaluate the following definite integral(s).

1. $\int_{-1}^2 x^2(x^3 + 1)^4 dx$

Let $u = x^3 + 1$ so $dx = \frac{du}{3x^2}$, $u(-1) = 0$, and $u(2) = 9$.

$$\begin{aligned} &= \int_0^9 x^2 u^4 \frac{du}{3x^2} \\ &= \frac{1}{3} u^4 \Big|_0^9 = \frac{9^4}{3} - 0 = \frac{9^4}{3} \\ &= \frac{6561}{3} = 2187 \end{aligned}$$

1. $\int_{-2}^1 \frac{t}{\sqrt{4-t^2}} dt$

Let $u = 4 - t^2$ so $dt = -\frac{du}{2t}$, $u(-2) = 0$, and $u(1) = 3$.

$$\begin{aligned} &= \int_0^3 t \cdot u^{-1/2} \frac{du}{-2t} \\ &= -\frac{1}{2} \left(\frac{2}{1} \right) u^{1/2} \Big|_0^3 = -\sqrt{3} \ (\approx 1.73205) \end{aligned}$$

A full 200 gasoline tank begins to leak out of at the rate of $50 - 2t$, where t is measured in minutes.

2. How much gasoline leaked out in the first 4 minutes?

$$\int_0^4 50 - 2t dt = \left(50t - \frac{2}{2}t^2 \right) \Big|_0^4 = 200 - 16 = 184 \text{ gallons}$$

The function $v(t) = -\sin(t)$, on $0 \leq t \leq 3/2\pi$, is the velocity in m/sec of a particle moving along the x -axis.

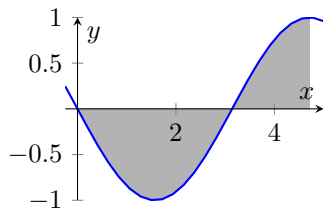
3. [2 points] If the particle has an initial position of $s(0) = 1$, find the function $s(t)$ giving its position.

$$\begin{aligned} s(t) &= -\int \sin(t) dt \\ &= \cos(t) + C, \quad s(0) = 1 \Rightarrow C = 0 \\ &= \cos(t) \end{aligned}$$

4. Find the total distance traveled over the given interval.

$-\sin(t) = 0$ on $[0, 3/2\pi]$ at $t = 0$ and $t = \pi$. So,

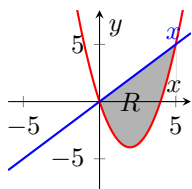
$$\begin{aligned} \text{distance} &= \int_0^{3/2\pi} |-\sin t| dt \\ &= \int_0^{\pi} \sin t dt + \int_{\pi}^{3/2\pi} \sin t dt \\ &= 2 + 1 = 3 \end{aligned}$$



Use calculus to find the area of the region described.

5. The region bounded by $y = x$ and $y = x^2 - 4x$.

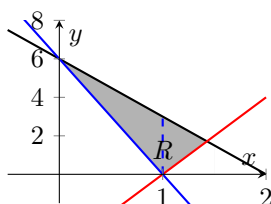
$$\begin{aligned} R &= \int_0^5 x - (x^2 - 4x) \, dx \\ &= \left. \frac{5}{2}x^2 - \frac{1}{3}x^3 \right|_0^5 = \frac{125}{6} \approx 20.8333 \end{aligned}$$



Find the definite integral(s) which give the area of the region described. **For this problem you only need to write the simplified integral which gives the area.**

6. The region in the first quadrant bounded by $-3x + 6$, $4x - 4$, and $-6x + 6$. Use calculus.

$$\begin{aligned} R &= \int_0^1 -3x + 6 - (-6x + 6) \, dx + \int_1^{3/2} -3x + 6 - (4x - 4) \, dx \\ &= \int_0^1 3x \, dx + \int_1^{3/2} -7x + 10 \, dx \end{aligned}$$



Use the general slicing method and to find the volume of the solid.

7. [8 points] The solid lies between planes perpendicular to the x -axis at $x = 0$ and $x = \pi/4$. The cross-sections are squares which are perpendicular to the x -axis and have a height of $\sqrt{\cos 2x}$ measured positively from the x -axis.

The cross-sections are square with sides of $\sqrt{\cos 2x}$. So then the area of each cross-section is $(\sqrt{\cos 2x})^2 = \cos 2x$.

$$\begin{aligned} V &= \int_0^{\pi/4} \cos 2x \, dx \\ &= \left. \frac{1}{2} \sin 2x \right|_0^{\pi/4} = \frac{1}{2}(1) - \frac{1}{2}0 = \frac{1}{2} \text{ cubic units} \end{aligned}$$

Let R be the region bounded by the following curves. Use the disk or washer method to find the volume of the solid generated when R is revolved about the x -axis.

8. $y = \sqrt{x}$, $x = 0$, and $x = 9$.

$$\begin{aligned} V &= \pi \int_0^9 (\sqrt{x})^2 dx \\ &= \pi \frac{1}{2} x^2 \Big|_0^9 = \frac{\pi 81}{2} \approx 127.235 \text{ cubic units} \end{aligned}$$

For this problem You only need to write the simplified integral which gives the volume.

9. $y = -16x$ and $y = 8x^2 + x^3$.
 $-16x = 8x^2 + x^3 \Rightarrow x(-x^2 - 8x - 16) = x(x+4)(x+4) = 0$. So $x = 0, -4$

$$\pi \int_{-4}^0 [(-16x)^2 - (8x^2 + x^3)^2] dx = \frac{32768}{35}$$

Find the length of the curve.

10. $y = 2x^{3/2}$ from $x = 0$ to $x = \frac{5}{4}$.

$$y' = 3x^{1/2} \text{ and } y'^2 = 9x.$$

$$\int_0^{5/4} \sqrt{1+9x} dx \text{ let } u = 1+9x. \text{ So then } dx = \frac{du}{9}.$$

$$\begin{aligned} L &= \frac{1}{9} \int_0^{5/4} u^{1/2} du \\ &= \frac{1}{9} \frac{2}{3} u^{3/2} \Big|_0^{5/4} = \frac{2}{27} \left(1 + 9 \left(\frac{5}{4} \right) \right)^{3/2} = \frac{335}{108} \end{aligned}$$

BONUS: Use calculus and the general slicing method to find a general formula for the volume of a pyramid with a square base.

Place the pyramid on the Cartesian plane such that the base is perpendicular to the y -axis and the peak touches the x -axis (as we did on the quiz). Let s be length of the base, and h be the height. The equation which gives the distance from the x -axis to the side of the pyramid is the line $y = -\frac{sx}{2h} + \frac{b}{2}$ (the line has slope $\frac{s/2 - 0}{0 - h}$ and y -intercept $(0, b/2)$). So the area of the cross-section is given by

$$A(x) = \left(2 \left(-\frac{sx}{2h} + \frac{s}{2} \right) \right)^2 = \left(\frac{s(h-x)}{h} \right)^2 = \frac{s^2}{h^2} (h-x)^2$$

and

$$\begin{aligned} V &= \int_0^h \frac{s^2}{h^2} (h-x)^2 dx = \frac{s^2}{h^2} \int_0^h (h-x)^2 dx \\ &= s^2 \frac{1}{3h^2} (h-x)^3 \Big|_0^h = s^2 \frac{h^3}{3h^2} = \frac{s^2 h}{3} \end{aligned}$$

The area of the base, $b = s^2$. So this is often written $V = \frac{1}{3}bh$.